Year 8

Term 1 Test 2019

Baldivis
Secondary College

Resources allowed: Calculator and a handwritten A4 notes page double sided

Name: Solutions Class: ______ Time: 50 minutes Mark:_____/46

CALCULATOR FREE

20 minutes

Mark:____/20

1) [6 marks]

Fill in the missing number in each of the following

a) $4 + (-7) =$ -3 ✓

b) $-3 - (-2) =$ -1 ✓

c) $(-4) \times (-8) =$ 32 ✓

d) $-16 \div 8 =$ -2 ✓

e) $(-4) \times 3 + 5 =$ -7 ✓

f) $-12 - 5 \times 4 =$ ~~-68~~ -32 ✓

2) [1, 1, 1, 1 = 4 marks]

Evaluate the following expressions, writing your answers as **basic numerals *(numbers)***:

a) 3^0 = 1 ✓

b) $\frac{5^3}{5^3}$ $5^0 = 1$ ✓

c) $3^2 + 3^0$

$9 + 1 = 10$ ✓

d) 2^5

4 8 16

$2 \times 2 \times 2 \times 2 \times 2 = 32$ ✓

3) [1, 1, 1 = 3 marks]

Write the complementary event for the following:

a) It will snow today

Not snow ✓

b) Rolling a 6-sided die and getting a 4

$\frac{1}{6} = \frac{5}{6}$ ✓ or

Not rolling a four ✓

c) You ride your bike home

Not ride your bike home ✓

4) [1, 1, 1, 2, 2 = 7marks]

Express the following in **simplest index form** using the index laws

a) $(7^3)^3$

$7^{3\times3}$

7^9 ✓

b) $\frac{5^7}{5^2} = 5^5$ ✓

c) $4^3 \times 4^4$

4^7 ✓

d) $\frac{6^4 \times 6^7}{6^5}$

$\frac{6^{11}}{6^5} = 6^6$ ✓ ✓

e) $\frac{(5^4)^3}{5^9}$

$\frac{5^{12}}{5^9} = 5^3$ ✓ ✓

Section 2: Calculator Assumed

30 minutes
Mark:____/26

Solutions

5) [3 marks]

Complete the following conversions:

a) 37cm = 0·37 m   ÷ 100 ✓

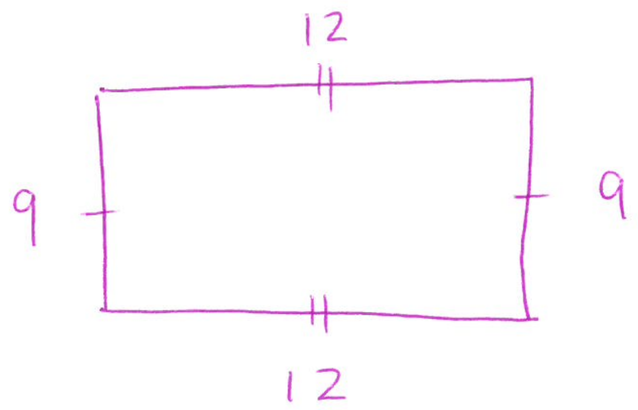
b) 63700m = 63·7 km   ÷ 1000 ✓

c) $7cm^2$ = 700 mm^2 $\times 10^2$ ✓

6) [1, 1 = 2 marks]

The perimeter of a rectangle is 42 m.

a) Draw a rectangle with this perimeter and mark in a possible length and width.



Or other combo's

✓ must include dimentions.

b) Find the area of your rectangle.

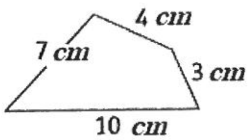
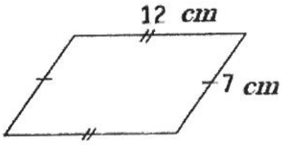
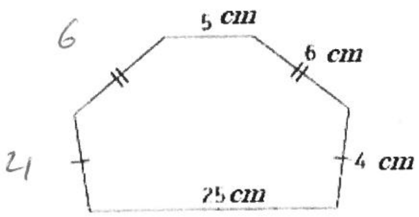
$9 \times 12 = 108m^2$

(½)   (½)

5

7) [1, 1, 1, + 1 mark for **all** units = 4 marks]

Find the **perimeter** of the following shapes. Show your working clearly and giving your answers correct to two decimal places when necessary:

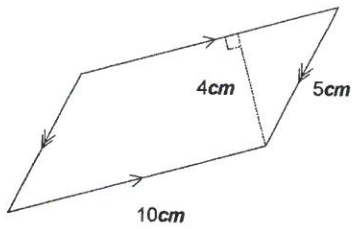
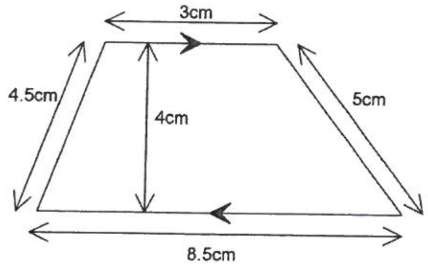
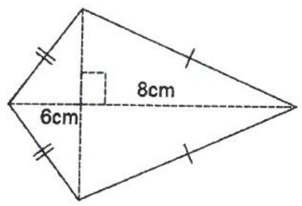
Shape	**Working and Answer**
a) 	7 + 4 + 3 + 10 = 24cm ✓
b) 	12 + 12 + 7 + 7 = 38cm ✓
c) 	4 + 6 + 5 + 6 + 4 + 25 = 50cm ✓

✓ units

4

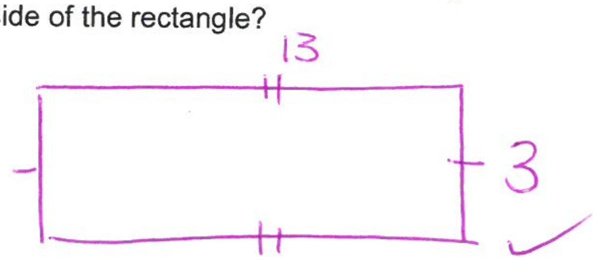
8) [1, 2, 1 = 4 marks]

Find the **area** of the following shapes. Show your working clearly and give your answers correct to two decimal places when necessary.

Shape	Working and Answer
a) 	4×10 $= 40cm^2$ ✓
b) 	$0.5 \times (3+8.5) \times 4$ ✓ $= 23cm^2$ ✓
c) 	$0.5 \times 6 \times 8$ $= 24cm^2$ ✓

9) [2 marks]

The area of a rectangle is $39cm^2$. If the length of the longest side is 13cm, what is the length of the shorter side of the rectangle?



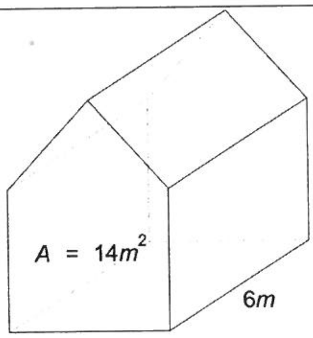
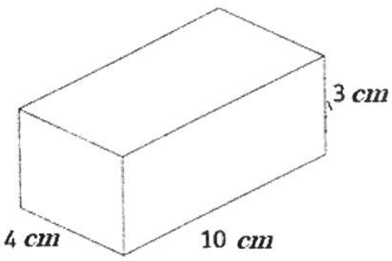
$39 \div 13 = 3$ ✓

diagram ✓
answer: 3 ✓

6

10)[2, 2 = 4 Marks]

Calculate the **volume** of this prism:

Shape	Working and Answer
	14×6 ✓ $= 84\ m^3$ ✓
	$4 \times 10 \times 3$ ✓ $= 120cm^3$ ✓

4

11) [3 marks]

There are 10 counters in a bag. The counters are numbered 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

If one counter is drawn out at random, what is the probability that it is a counter:

a) with an odd number

$\frac{5}{10}$ ~~$\frac{1}{2}$~~ ✓

b) that is greater than 6

$4/10$ ✓

c) that is not greater than 6

$6/10$ ✓

12) [2 mark]

a) If the probability of selecting a blue marble from a bag of marbles is 0.3. Find P (Not Blue)

$\cancel{0}$ $1 - 0.3 = 0.7$ ✓

b) If the probability of not selecting a red marble from a bag of marbles is $5/11$. Find P(Red)

$\frac{11}{11} - \frac{5}{11} = \frac{6}{11}$ ✓

13) [1, 1 = 2 marks]

If we write the letters from the word MATHEMATICS on separate pieces of paper and place them in a bag, the probability of selecting an M is: $P(M) = \frac{2}{11}$.

a) Write down a word which has $P(S) = \frac{1}{4}$.

~~sok~~ soak ✓

b) Write down a word which has P(vowel) > P(consonant).

eat ✓

End of Test

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